



Sigvard Bratlie

Data Scientist

M.Sc. student in Data Science at NMBU with 4 years of project management experience in commercial real estate. Currently researching agentic AI for the legal sector.

Timeline

Data Science (M.sc.) | NMBU

Aug 2024 - Present

- Researching context preservation in multi-turn LLM conversations for legal case management at Bahr AS for my master thesis — building a LangGraph-based agent that maintains structured case state across extended conversations, with RAG on Norwegian law, automated document analysis, and multi-LLM support. Evaluating whether explicit state management improves accuracy over baseline RAG as conversation length increases.
- Built a YOLO11 object detection model for Rumex identification in drone imagery. Also worked with CNNs, canny edge detection, and classical image processing techniques.
- Built an interactive Streamlit dashboard analyzing Norwegian electricity production/consumption vs. historical weather data, with time-series decomposition, forecasting, and outlier detection.
- Developed an end-to-end ETL pipeline scraping Finn.no, enriching with public registry data (Geonorge, Statens Vegvesen, Kartverket), and training pricing models with XGBoost/CatBoost/LightGBM on GCP.
- Built SIBR Market — an AI valuation platform where users chat with LLM agents for real estate and vehicle valuations, deployed on Google Cloud Run.
- Built an oil spill diffusion simulator with mesh-grid modeling and auto-generated MP4 visualizations.
- Statistical theory (Bayesian/frequentist, MLE, multivariate distributions), optimization (gradient descent, Newton's method, KKT, simplex), deep learning, and algorithm

design.

Project Manager | Newsec

Aug 2020 - Jul 2024

- Managed multi-million NOK construction and tenant fit-out projects end-to-end, coordinating 20-30 stakeholders across consulting engineers, contractors, architects, clients, and municipal authorities. Portfolio included residential conversions, commercial rehabilitations, and public space developments across central Oslo.
- Delivered projects for institutional property owners and international investors — including OBOS, Pareto Securities, Clarkson Platou, SBB Norden, and Selvaag — handling both new developments and commercial tenant adaptations.
- Led technical due diligence on property transactions across office, retail, industrial, and residential assets, producing integrated assessments for both buy-side and sell-side clients.
- Managed full procurement cycles — tender processes, bid evaluations, contract negotiation, and administration under NS8407/NS8402 frameworks — including public sector procedures and regulatory applications.
- Owned project financial reporting and ensured compliance with regulatory frameworks and sustainability standards including BREEAM.

Engineering (B.sc.) | NTNU

Aug 2017 - Jun 2020

- B.Sc. in Structural Engineering with focus on construction management, structural design, and regulatory frameworks.
- Completed a feasibility study for the redevelopment of Mustad Industrial Park as part of a real estate development course.
- Bachelor thesis on replacing structural steel systems with timber — analyzing environmental impact, cost implications, and structural performance.
- Foundation in project planning, contract management, and navigating building regulations and public approval processes.

Additional Experience

- Continued taking on advisory and project management assignments from former clients on an hourly basis alongside studies.
- Economics coursework at BI Norwegian Business School (2021-2022).

Skills

Machine Learning • ANN (PyTorch & TensorFlow) • Transformers (HuggingFace) • ML (Scikit Learn, XGBoost, ++) • Agents (Langchain/langgraph) • Data handling (Pandas, NumPy, Spark) • Plotting (Matplotlib, Seaborn, Plotly) • Computer vision (OpenCV, YOLO)	Cloud Computing • GCP, Azure & AWS • Data warehouse (Big Query, Snowflake) • CI/CD tools (Cloud Build, etc.) • Serverless computing (Cloud Run) • Docker • VM and Kubernetes • NoSQL DB (MongoDB, Firestore) • ++ (Secrets, IAM handling, etc.)	Other Tools/Skills & Professional • FastAPI • Git • Building SDK • Streamlit • Dashboards (PowerBI, Looker) • Figma • Project Management • Client Relations • Business Acumen • Reporting & Presentation
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Projects

- [Agentic Legal AI \(Master Thesis\)](#): LLM/NLP project for legal document processing.
- [Weather & Electricity Dashboard](#): Data analysis and EDA dashboard.
- [Rumex Detection Model](#): Computer vision model using YOLO.
- [Market Data ETL Project](#): Data fetching and scraping pipeline.
- [SIBR Market](#): UI/LLM-based market interface.

Contact Info

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 LINKEDIN

 GITHUB

 EMAIL